AI in Education – Automated News Briefing Dashboard

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Github Repository:

https://github.com/AsmaeNakib/AI-in-Education-Automated-News-Briefing-Dashboard

Introduction

Artificial intelligence (AI) is rapidly transforming education, bringing both revolutionary innovations and serious concerns. Recent surveys show that over 85% of teachers have already experimented with AI tools in the classroom, often without prior formal training. Policymakers are increasingly concerned about issues such as data privacy, academic integrity, and algorithmic bias. As AI becomes more integrated into the learning process—from intelligent tutoring systems to automated grading—its impact on students and schools is also growing. Therefore, it is essential that educators, school leaders, and policymakers stay informed about the evolving role of AI in education. However, keeping up with the daily developments in the field of AI in education has become challenging. News and research on the topic are scattered across technology blogs, educational journals, political reports, and the mainstream media. Manually monitoring this information overload and distinguishing between credible news and misinformation is time-consuming and prone to error.

This project addresses this challenge by automatically tracking news trends in the field of AI in education using AI. We have developed an automated news dashboard – an AI-powered system that collects and summarizes the latest news on AI in education daily. The dashboard aims to efficiently and continuously inform key stakeholders (teachers, school administrators, education policymakers, researchers) about important developments. By automatically collecting and summarizing news, the tool provides a daily overview of the latest developments in this area. Stakeholders no longer need to search dozens of websites but receive a single, comprehensive update daily, saving time and reducing information overload.

The following sections describe how the dashboard was developed and how it works, how we verified its accuracy, what challenges we overcame, and what opportunities exist for the future to further expand this approach.

Project Development Process

We implemented the system using n8n, a workflow automation tool that allowed us to connect data sources, transformation logic, and AI services. The workflow follows a clear sequence of steps: from raw data collection to filtering and summarization to delivery of the final dashboard, which we explain in more detail below. The key phases are: (1) Data collection via news feeds, (2) content cleansing and filtering, (3) data aggregation and formatting, (4) AI summarization using a large language model (LLM), and (5) dashboard creation and delivery. By chaining these steps together, the system automatically converts a series of news articles into a concise summary on a daily basis.

Data Collection (News Feeds)

The process begins with data collection from online news sources. We used Google News RSS feeds to automatically retrieve the latest articles on AI in education. Specifically, two specific RSS queries (Google News searches) were performed: one for news on "AI in education" and another for news on "EdTech AI", to gather recent articles on artificial intelligence in the educational context. Using the RSS reader nodes integrated into n8n, the workflow retrieves a list of recent headlines, summaries, publication dates, and links for each relevant article. This approach ensures broad coverage of topics and sources, as Google News aggregates articles from various publishers worldwide. By including both general news on "AI in education" and more specialized news on "EdTech AI," we capture developments in primary and secondary schools, higher education, educational technology companies, policy announcements, and research findings. The result of the data collection is a series of article entries (usually several dozen per day) with basic metadata (title, source, summary, URL, timestamp) that are ready for further processing.

Filtering and Pre-Processing

After collecting the raw data from the news articles, the next step was to clean and standardize the content. Since RSS feeds often contain duplicates, we added filter logic in n8n using JavaScript and field editing nodes to clean up the data. Each article was cleaned up to remove HTML artifacts and unnecessary tracking information in the URLs, and we ensured that the fields were consistent. We also extracted the source domain from each link so we could see where the articles came from (e.g., techcrunch.com or educationweek.org). To stay focused, we performed simple keyword checks to filter out any articles that might not be relevant to the topic. At the end of this process, we had two well-structured lists of articles (one from each feed) ready to be merged.

Data Aggregation and LLM Preparation

After cleaning each feed, the workflow aggregates the data. The two article streams (from the "AI Education" RSS feed and the "EdTech AI" RSS feed) are merged into a single, unified collection. We used an n8n merge node for this. Now we have a single set of all unique AI-related news articles on education for the given day. This merged dataset serves as input for the language model. Preparation involves converting the structured article data into a long block of text that the language model can process. This structured document containing multiple articles allows the language model to process all of the day's news with a single request.

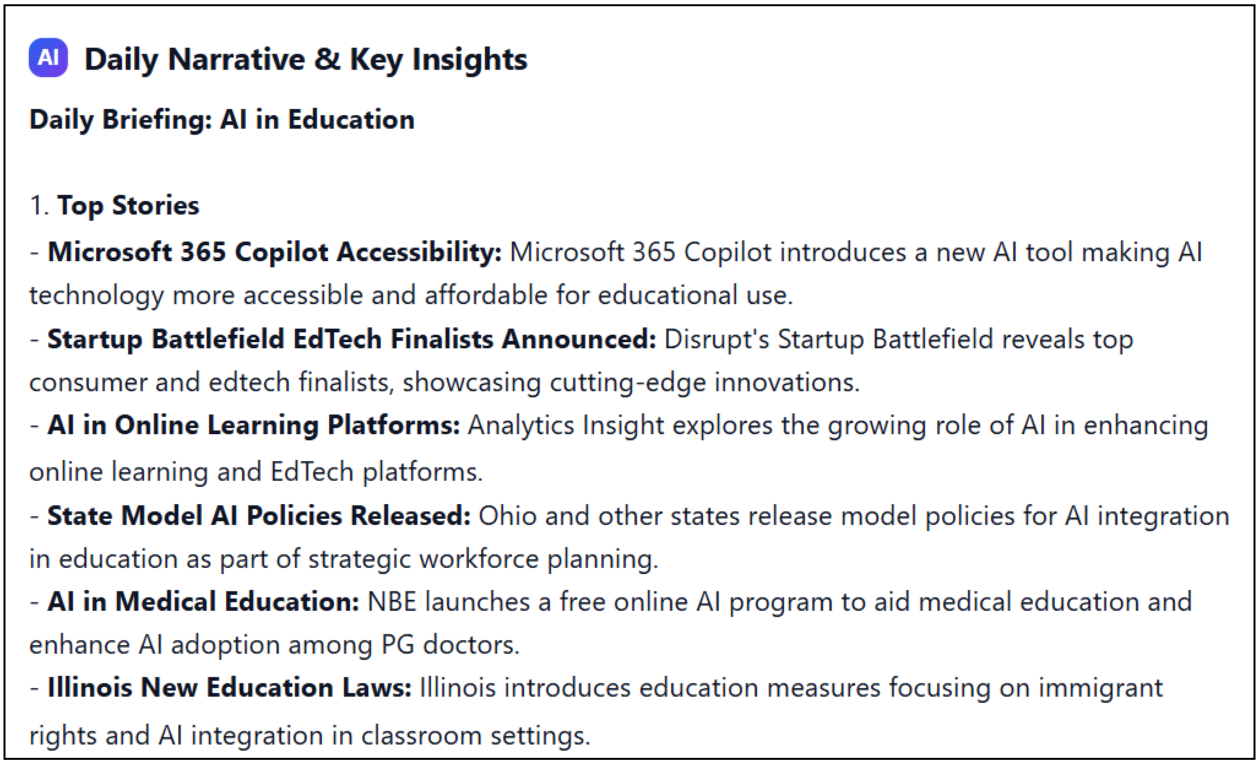
AI Summarization and Trend Extraction

After the day's articles were collected and formatted, the AI summary forms the core of the system. We used OpenAI's large language model GPT-4 (via n8n's OpenAI API integration) to analyze the collected news texts and generate a structured summary of the key findings. GPT-4 is instructed to identify the day's top stories (usually 5–7) and summarize each in one or two sentences. The model is also tasked with identifying overarching trends and patterns across the articles and noting any risks or challenges mentioned (e.g., ethical concerns, controversies, or issues highlighted in the news). Additionally, the AI is asked to suggest opportunities or positive developments (such as new initiatives, successes, or promising research in AI in education). Finally, we request a set of trending keywords or topics, as well as a brief overview of the sources (e.g., which media outlets published multiple articles). The choice of GPT-4 is based on its ability to understand context and produce coherent, well-structured texts.

The model's output is in a structured text format (JSON) that can be processed by our workflow. Given the educational context, it was important for us to design the task neutrally and informatively, the AI was explicitly instructed to avoid sensationalist language and to limit itself to factual, concise summaries.

Dashboard Generation and Delivery

The final step in the pipeline is to generate the dashboard interface and make it available to users. Once the GPT-4 model returns the structured briefing content, the n8n workflow analyzes the output and converts it into a formatted HTML page. In this project, we designed the dashboard as a simple web page (accessible via a public n8n webhook URL) that presents the AI summary in a visually clear format. A custom formatting node combines the various parts (top stories, trend analysis, risks, opportunities, etc.) into sections with headings, bullet points, and highlights. For example, "top stories" could be displayed as a numbered list of headings (each linked to the original article), accompanied by a one-line summary.

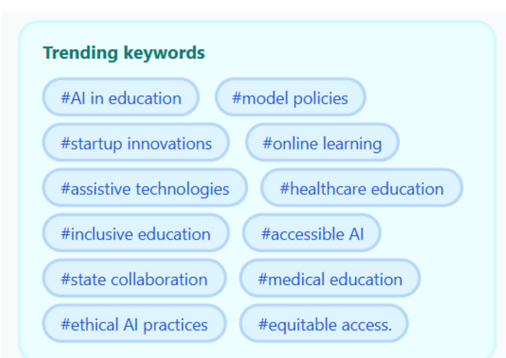


AI **Daily Narrative & Key Insights**

Daily Briefing: AI in Education

1. **Top Stories**
- **Microsoft 365 Copilot Accessibility:** Microsoft 365 Copilot introduces a new AI tool making AI technology more accessible and affordable for educational use.
- **Startup Battlefield EdTech Finalists Announced:** Disrupt's Startup Battlefield reveals top consumer and edtech finalists, showcasing cutting-edge innovations.
- **AI in Online Learning Platforms:** Analytics Insight explores the growing role of AI in enhancing online learning and EdTech platforms.
- **State Model AI Policies Released:** Ohio and other states release model policies for AI integration in education as part of strategic workforce planning.
- **AI in Medical Education:** NBE launches a free online AI program to aid medical education and enhance AI adoption among PG doctors.
- **Illinois New Education Laws:** Illinois introduces education measures focusing on immigrant rights and AI integration in classroom settings.

Figure 1: Dashboard Top Stories section



The "Trending Keywords" might be displayed as small badges or a comma-separated list. We also overlay stylistic elements such as colored cards or icons for visual appeal.

Figure 2: Dashboard Trending Keywords section

2. **Trends, Insights & Patterns**
- **Policy Development:** There is a significant trend of states developing model policies to guide AI integration in education.
- **Startup Innovations:** New startups continue to push boundaries in edtech, with focus on AI-driven tools and platforms.
- **Inclusive Technology:** Growing emphasis on AI-enabled assistive technologies designed to support inclusive education.

Figure 3: Dashboard Trends, Insights & Patterns section

3. **Risks, Challenges & Concerns**
- **Safety and Ethics in AI Adoption:** The need for safe AI practices in educational settings is a recurring concern across articles.
- **Equity and Access:** Discussions highlight potential disparities in AI access, urging equitable resource distribution.

Figure 4: Dashboard Risks, Challenges & Concerns section

4. **Opportunities & Promising Practices**
- **Collaboration on AI Models:** States collaborating on AI policies provide a framework for shared learning and consistency in AI-education integration.
- **Assistive Technology Implementation:** AI tools begin to transform inclusive education, accommodating diverse learning needs effectively.
- **Medical Education Advancements:** Free AI programs for medical practitioners encourage safe, widespread AI adoption in healthcare education.

Figure 5: Dashboard Opportunities & Promising Practices section

To deliver the dashboard, the workflow uses an HTTP webhook response that outputs the HTML content. In practice, this means that the automated workflow (either through a daily schedule or a manual request) performs all steps and finally returns the compiled HTML page as a response.

The dashboard can thus be accessed at any time via a web browser using the webhook URL and always displays the latest daily overview. We chose this deployment method for simplicity, as no separate web server is required because n8n hosts the page via its webhook. Furthermore, the design could easily be extended to other channels: for example, we could email the HTML file with the briefing to subscribers every morning or integrate it into a learning management system for educators. The most important result of this phase is a sophisticated, interactive dashboard that any user (with the link) can access to receive their daily news on AI in education. Automation is end-to-end: no manual intervention is required from data collection to the public dashboard. However, as explained below, we have taken measures to verify the credibility of the content and monitor the performance of the system over time.

Verification and Results

Ensuring the credibility and accuracy of an automated news summary is of utmost importance, especially in an educational context where users must trust the information. Throughout development and deployment, we used various methods to verify the dashboard output.

First, we ensured that the summary always cites its sources. All sources are listed at the end of the dashboard so readers can easily check the full article to understand the context or confirm information.

News feeds monitored

Current Google News queries powering this dashboard

Source	# Articles
- edtechmagazine.com	1
- techcrunch.com	1
- findarticles.com	1
- analyticsinsight.net	1
- yahoo.com	1
- nprillinois.org	1
- spectrumnews1.com	1
- capitolnewsillinois.com	1
- amritavidya.ai	1
- medicaldialogues.in	2
- indiatoday.in	1

Figure 6: Dashboard Sources

In user tests with some colleagues, this feature was appreciated because it gave users the assurance that they could always trace a summary back to a reputable source. It also provided an implicit accuracy check: if the AI summary misinterpreted a story, a reader comparing it to the original would notice the discrepancy. In our evaluations, we did not encounter any gross misinterpretations or factual errors in the summaries.

Another aspect of verification was daily monitoring by the project team. Although the process is automated, during the pilot phase we checked the dashboard results daily to ensure they were plausible and consistent with actual news stories. This informal review served as a kind of "human-in-the-loop" plausibility check.

CSV exports of the collected news data provide a layer of auditability to verify the reliability of the system. Each day's CSV file contains titles, summaries, and timestamps of articles collected within a 24-hour period. Comparing consecutive daily CSV files shows that the pipeline is updated with new content every day rather than reusing old articles. The articles differ from day to day, confirming that date-based filtering is working correctly to capture new publications in each cycle. At the same time, there is continuity in the overarching themes (for example, topics such as AI in education or governance appear over several days), suggesting that the system consistently tracks relevant news categories while reflecting the current discourse of each day.

Manual review of these CSV files also confirms the quality and integrity of the data fed into the AI summarizer. Each entry has a descriptive title and meaningful summary, and obviously low-quality or irrelevant items were filtered out before reaching the AI. The system's duplicate removal measures are also effective. No identical article appears twice in the CSV file for the same day, so each daily briefing consists of unique items (although similar articles from different sources may still appear on different days, which is a known limitation). One known limitation is that the article URLs in the CSV file are redirect links from Google News rather than direct publisher URLs, which makes it difficult to analyze source diversity.

We recognize that ongoing monitoring will be important, as the AI model or news sources evolve, the prompts and filters may need to be further refined. In summary, reviewing the dashboard was an ongoing process that combined automated safeguards (source attribution, use of multiple sources) with human oversight (daily review and user feedback).

Risks and Challenges of AI in Education

The use of an AI-driven tool in education required addressing several overarching risks and challenges. Many of these are not only relevant to our project, but reflect general concerns about the use of AI in education, including data privacy, algorithmic bias, over-automation (lack of human control), loss of context, and environmental impact.

Data Privacy and Security

An important concern in relation to AI in education is data protection. Schools and educational institutions process sensitive data from students and staff, so any AI integration must protect this information (Vilcarino & Langreo, 2025). In our news dashboard project, the data entered and generated consisted of public news articles, so we did not directly handle any personal data from students. Nevertheless, we based our approach on general data protection principles. When using external AI APIs (such as GPT-4 from OpenAI), for example, it is important to consider what data is sent to these services. We limited the inputs to public texts (no confidential information) and followed the usage guidelines to ensure that no private data was accidentally disclosed. We make it clear that the dashboard only uses publicly available information.

Algorithmic Bias and Representation

Algorithmic biases are a well-documented risk in AI systems and can manifest themselves in concerning ways in educational AI (UNESCO, 2021). In a news aggregator like ours, biases can arise in various ways: in the selection of sources, in the filter criteria, or through biases learned by the language model itself during summarization. We recognized the risk that the dashboard could systematically favor certain perspectives or topics. For example, if our news feeds or sources were biased toward news from the tech industry, critical or skeptical viewpoints on AI in education might be underrepresented in the briefings (Vilcarino & Langreo, 2025).

Another initial limitation was that our sources were primarily English-language and US-focused, meaning we could miss important developments from non-English-speaking regions, an issue of equity in the global education context. This type of representational bias could convey an incomplete or one-sided picture of AI trends in education (UNESCO, 2021).

To mitigate these issues, we took deliberate measures. We tried to include a variety of different sources in the feed (as far as this was possible with Google News search): These included mainstream media, niche websites for education technology, and even official press releases on policy measures or international organizations. Of course, our selection of sources is not exhaustive and is subject to a certain degree of bias (based on what we consider to be reputable sources), so this is an area where we need to continuously improve. Our approach is consistent with new guidelines that emphasize fairness in educational AI: International frameworks such as UNESCO's AI Ethics Recommendation call for measures to prevent algorithmic bias and ensure fair representation of different cultures and viewpoints.

Over-Automation and Need for Human Oversight

A key question in projects like this is the extent to which one should rely on automation or human expertise (UNESCO, 2021). Excessive automation such as leaving all decisions to AI without human oversight, can be risky, especially in education, where the stakes are high (learning outcomes, public trust). Throughout our process, we ensured that at least one of us regularly reviews the results, and in a real-world application, an editor or moderator would review the daily briefing before it is distributed.

There are several reasons why human oversight is essential. One problem is the possibility of misinformation. If an article from a questionable source slips through, or if a news item itself is inaccurate, the AI could summarize it in good faith, and the dashboard would end up disseminating false or misleading information (Vilcarino & Langreo, 2025). While we have built safeguards into our workflow by primarily using reputable sources and keywords, no filter is foolproof.

Another reason is quality control: humans can spot when the AI summary has omitted an important detail or when a key piece of news is missing from the briefing, and can then adjust the system accordingly.

Environmental Sustainability

Environmental impact is one of the most often overlooked challenges when using artificial intelligence. Running large AI models, even for simple tasks like summarizing text, consumes computing resources and electricity. In an era where educational institutions are increasingly focused on sustainability, the introduction of AI tools should consider their carbon footprint. During this project, we observed that running a powerful model like GPT-4 daily consumes significant energy. While each individual summary is relatively small, our estimates suggest that a single query from a GPT-3/4 of this size might consume only about 0.3 watt-hours of energy. These costs accumulate as the model's size increases. If many schools or users run similar AI operations daily, or if the model has to process thousands of queries, the cumulative energy consumption can become substantial. Recent research indicates that millions of AI queries per day can collectively approach the energy consumption required to train large models if run continuously (UNESCO, 2021).

We have taken some concrete steps to address this issue. The workflow is scheduled to run only once a day, rather than hourly or on demand, to avoid excessive computation. We have also limited the number of articles and the length of the prompts processed.

Opportunities and Future Directions

This dashboard project, while a standalone prototype, offers numerous opportunities for further use and development. We see promising potential in expanding its practical applications and enhancing its technical capabilities. Below, we outline some key opportunities for the future application, expansion, and improvement of an AI-powered news feed dashboard.

Wider Adoption in Educational Institutions and Policy Circles: The next step is to deploy tools like this dashboard in real-world teaching and research settings. Our project has highlighted, in particular, the potential for collaboration with institutions to tailor the brief to their needs, such as highlighting news in local languages for regional officials. This suggests multilingual support: one opportunity is to integrate non-English news sources or use translation so that the dashboard is not limited to English sources. In addition, researchers in educational technology or sociology can use the dashboard to track emerging trends and research findings in artificial intelligence and education.

Technical Enhancements – Custom Queries and Interactivity: On a technical level, there are many ways to expand the dashboard's capabilities. One idea is to enable users to personalize and interact with it. Currently, the dashboard provides a fixed daily summary. In the future, we could allow users to ask follow-up questions to an AI system about the news. For example, after reading the daily summary, a teacher could ask, "Give me more details about the AI tool used in the second news item," and the integrated learning management system would generate a detailed explanation or even fetch additional sources.

Another improvement would be support for custom queries or filters—a user could specifically subscribe to AI in education news about "data privacy" or "STEM education," and the system would personalize the summary accordingly. Technically, this could be achieved by maintaining multiple sources for topics or by dynamically querying news APIs based on user keywords.

Each of these additions will require careful formatting and potentially more sophisticated AI.

Continuous Improvement – Feedback and Editorial Integration: As we've emphasized previously, continuous improvement through human oversight and feedback is one of the most important factors for the success of any AI-powered education system. Looking ahead, we envision integrating a more formal mechanism for receiving end-user feedback. Users could report any summary they find misleading or unhelpful, or rate the relevance of the daily summary. This feedback could then be used to refine the system.

In conclusion, the AI in Education Automated News Feed Dashboard has achieved its immediate goal: providing concise daily updates on the intersection of AI and education through an automated system. More importantly, it has demonstrated how AI can be harnessed to manage the massive flow of information in a critical field, highlighting the need for thoughtful application. The dashboard has scalability: from customization features to wider deployment, and each expansion will be guided by the lessons learned about diverse input, ethical transparency, and the invaluable contribution of human insights. Building on this foundation, educational institutions and communities can stay informed about AI in an innovative and trustworthy way. As the conversation around AI in education continues to evolve, we see this project as a step toward ensuring stakeholders have easy access to the latest knowledge, viewing AI as a helpful ally rather than a simple solution. The opportunity now lies in improving and scaling these tools, fostering a learning environment where keeping pace with technological changes is less burdensome and more collaborative and intelligent.

Appendices:

Appendix A: GitHub Repository

https://github.com/AsmaeNakib/AI-in-Education-Automated-News-Briefing-Dashboard

This repository includes the following deliverables:

- Dashboard Output - Dashboard.html The generated HTML dashboard for one execution of the system, presenting AI-generated summaries, trends, and monitored sources in a structured visual format.
- Automation Pipeline - PIPELINE.json The exported n8n workflow file defining the full automation pipeline, including data collection, filtering, aggregation, AI summarization, and dashboard generation.
- Sample Dataset - articles.csv A CSV export of the articles collected during one daily run of the system. This file is used in the report to support verification and validation of the data pipeline, including checks for freshness, relevance, and duplication.

Bibliography:

1. Jennifer Vilcarino & Lauraine Langreo (2025). **“Rising Use of AI in Schools Comes With Big Downsides for Students.”** *Education Week*, Oct 8, 2025. (Highlights concerns over AI’s negative impacts such as privacy breaches and bias in K-12 education).

2. UNESCO (2021). **Recommendation on the Ethics of Artificial Intelligence.** (Global framework emphasizing transparency, fairness, and human oversight in AI, including educational contexts).